ACTUARIAL RESEARCH · UNIVERSITY OF NAIROBI

Coverage Gaps in SHIF

Kenya's Informal Sector & the Cost of a Broken Premium Floor

Waweru B. Njihia et al. · Homa Bay County Case Study · 8 yrs NHIF data (2016–2024)

27.8%
of disposable income consumed by Ksh 300 floor (Ksh 5,000 earner)

98.2%
default probability for lowest-income informal workers

143.5%
projected Year 1 loss ratio under current SHIF structure

Executive Summary

Kenya's transition from NHIF to SHIF represents a historic expansion of mandatory health insurance — but the current pricing architecture creates a structural trap for informal-sector households. This research applies binomial logistic regression and ARIMA time-series forecasting to 8 years of historical NHIF financial data (2016–2024), using Homa Bay County as a proxy for Kenya's volatile agricultural informal economy (28.2% poverty rate; 83.6% informal employment). The central finding: the rigid Ksh 300 floor is regressive by design and guarantees immediate fund insolvency without structural subsidies.

1. The Affordability Crisis

The Statutory Floor Problem

The SHIF mandates a 2.75% income contribution with a Ksh 300 minimum: $\mathbf{C_i = max(0.0275 \times Y_i, 300)}$. This creates two critical inflection points:

Income Tier	SHIF Premium	Disposable Income	Affordability Burden	Status
Ksh 5,000	Ksh 300 (floor)	~Ksh 1,080	27.8%	CATASTROPHIC
Ksh 8,000	Ksh 300 (floor)	~Ksh 1,734	17.3%	CATASTROPHIC
Ksh 10,909	Ksh 300 (floor)	~Ksh 2,360	12.7%	ABOVE THRESHOLD
Ksh 15,000	Ksh 412.50	~Ksh 3,250	12.7%	ABOVE THRESHOLD
Ksh 18,182	Ksh 500	~Ksh 3,927	12.7%	NHIF BREAKEVEN
Ksh 25,000	Ksh 687.50	~Ksh 5,400	12.7%	STABLE

WHO catastrophic health expenditure threshold: 10% of total income or 40% of non-subsistence income. Every informal earner in Homa Bay surpasses this threshold under the current structure.

Scenario Stress Testing — Household Wallet

Scenario	Starting Income	Shock Applied	New Disp. Income	Default Probability
Baseline	Ksh 8,000	None	Ksh 1,734	98.2%
Inflation Shock	Ksh 8,000	+10% cost of living	Ksh 960	**100% (guaranteed)**
Agricultural Drought	Ksh 8,000	-25% income	Ksh 0	**100% (guaranteed)**

2. The Default Risk Model

A binomial logistic regression was executed in R on n=1,000 simulated Homa Bay households. All predictors are statistically significant at the 99.9% confidence level ($p < 0.001$). McFadden's Pseudo $R^2 = 0.3363$ (excellent model fit).

Variable	Coefficient (β)	Z-Value	p-Value	Risk Assessment
Intercept	-3.786	-4.589	4.46e-06	Baseline risk
Affordability Ratio (Ai)	+14.25	6.938	3.98e-12	**HIGH RISK ★★★**
Income Volatility (Vi)	+0.5818	5.937	2.91e-09	**HIGH RISK ★★★**
Absolute Gross Income (Yi)	-0.000142	-3.898	9.71e-05	**HIGH RISK ★★★**

3. Fund Sustainability — 5-Year Projection

Using ARIMA(0,1,0) time-series forecasting on 2016–2024 NHIF historical ledgers, the fund was stress-tested against two shocks inherent to informal-sector integration: a 25% premium attrition drop (mass default) and a 15% claim volatility spike (moral hazard).

Financial Metric	Baseline SHIF	Proposed APPM	Status
Total Contributions	Ksh 54,711.83M	Ksh 76,420.50M	
Total Liabilities	Ksh 78,545.69M	Ksh 74,100.20M	
Operating Surplus/Deficit	Ksh -23,833.86M	+Ksh 2,320.30M	
Year 1 Loss Ratio	**143.5% — FAILS**	**96.9% — PASSES**	
Actuarial Status	**TECHNICAL INSOLVENCY**	**SOLVENCY RESTORED**	

> **Technical insolvency in Year 1:** The fund starts with Ksh 4,648.5M in reserves. The Year 1 operating deficit of Ksh -23,833.86M instantly eliminates those reserves, plunging the fund into a structural deficit of approximately Ksh -19,185.36M. This manifests as unpaid hospital claims, drug shortages, and emergency exchequer bailouts.

4. The Fix: Adjusted Progressive Premium Model (APPM)

The APPM replaces the rigid Ksh 300 floor with a dynamically scaled formula: $\mathbf{C_i = (Y_i \times R_t) - S_i}$, where R_t is a tiered progressive rate and S_i is a state solidarity subsidy triggered automatically for households below the poverty threshold.

Tier	Income Range	Rate (Rt)	Subsidy (Si)	Out-of-Pocket
Tier 1	Below Ksh 10,909	1.0%	**100% subsidised**	**Ksh 0**
Tier 2	Ksh 10,909 – 19,999	1.5%	Partial	Scaled
Tier 3	Ksh 20,000 – 49,999	2.0%	None	Scaled
Tier 4	Ksh 50,000+	2.75%	None	Standard

Impact on a Ksh 5,000 earner: default probability drops from 98.2% → 5.1% (–93.1 pp). Mechanism: neutralising the Affordability Ratio (A_i) eliminates the dominant +14.25 coefficient from the logistic regression, collapsing the default driver.

5. Actionable Recommendations

1. Adopt tiered pricing and targeted subsidies

Scrap the Ksh 300 floor for earners below Ksh 10,909. Implement a socio-economic categorisation database (Rwanda's Ubudehe model) to trigger automatic state subsidies for the poorest.

2. Leverage Microfinance Institutions (MFIs)

Informal workers in MFIs are 44% more likely to maintain insurance. Integrate premium collection with local Chamas and table-banking groups to buffer seasonal dry periods.

3. Remove restrictive waiting periods

The current 90-day lockout for defaulters destroys Expected Utility. Benefits must be immediate upon contribution — otherwise rational actors will continue to default.

4. Diversify the revenue base

Reduce reliance on volatile out-of-pocket premiums. Shift lower-tier funding to indirect consumption-based taxes (Ghana model) to stabilise revenue against informal-sector income shocks.